

Basic Protocol of Rótuloselectronicos.net LED Display Controller

1.1 Data Packet Format

The data packet format of RS232/RS485

Data	Value	Length(Byte)	Description
Start code	0xa5	1	The start of a packet
Packet type	0x68/0xE8	1	0x68: Send packet 0xE8: Return packet
Card type	0x32	1	Fixed Type Code
Card ID	0x01~0xFE 0xFF	1	Control card ID, the screen No, valid values are as follows: 1 ~ 254: the specified card ID 0xFF: that group address, unconditionally receiving data
Protocol code		1	Protocol identification. Detail in the following "Command list"
Additional information/ confirmation mark	0 or 1	1	The meaning of bytes in the packet is sent, "Additional Information", is a packet plus instructions, and now only use the lowest: bit 0: whether to return a confirmation, 1 to return and 0 not to return bit 1 ~ bit 7: reserved, set to 0
Packet data	CC ○ ○ ○ ○ ○ ○		Packet data
Packet data checksum	0x0000~0xffff	2	Two bytes checksum. Lower byte in the former. The sum of each byte from "Packet type " to " Packet data" content
End code	0xae	1	The end of a packet (Package tail)

Command list

Command	Code	Description
Restart hardware	0x2d	
Restart APP	0xfe	
Write file(Open)	0x30	
Write file (Write)	0x32	
Write file (Close)	0x33	
Quick write file (Open)	0x50	
Quick write file (Write)	0x51	
Quick write file (Close)	0x52	
Time query and set	0x47	
Brightness query and set	0x46	
Query version info	0x2e	
Power ON/OFF info	0x45	
Power ON/OFF control	0x76	
Query temperature	0x75	
Remove file	0x2c	
Query free disk space	0x29	

Transcoding description :

The following process is done sending and receiving low-level functions, If you write your own PC side of the sending and receiving programs, you must implementation as below conventions. Use the without transcoding code to calculation checksum.

Send :

Between start code and end code, if there is 0xA5, 0xAA or 0xAE, it should be converted to two code :

0xa5 → 0xaa 0x05. The purpose is to avoid the same with the start character 0xa5

0xae → 0xaa 0x0e. The purpose is to avoid the same with the end of the symbol 0xae.

0xaa → 0xaa 0x0a. The purpose is to avoid the same with the escape character 0xaa.

Receive :

Received symbol 0xa5, said that the beginning of a packet

Received symbol 0xae, said that the end of a packet

When PC receive data from controller, if there is 0xA5, 0xAA or 0xAE, it should convert two code to one code, specifically for

0xaa 0x05 → 0xa5

0xaa 0x0e → 0xae

0xaa 0x0a → 0xaa

1.2 Packet Data

1.2.1 Restart hardware (CMD = 0x2d)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x00	1	Filling a byte value of 0x00

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.2 Restart APP (CMD = 0xfe)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Restart confirmation	4	"APP!"

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.3 Interactive upload file

1.2.3.1 Open file (CMD = 0x30)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Read/Write mode	1	'r': Read 'w': Write
0x0001	Read/Write type	1	'b': binary 't': Text mode (Now don't support)
0x0002	Hour	1	File time "Hour"
0x0003	Minute	1	File time "Minute"
0x0004	Second	1	File time "Second"
0x0005	Year	1	File time "Year", 2-digit number. For example 0x09 means 2009
0x0006	Month	1	File time "Month"
0x0007	Day	1	File time "Day"
0x0008	Reserved	1	Fill 0
0x0009	File size	4	File size. Low byte first.
0x000d	File name and file extensions	Variable-length	For example: "abc.txt"
Variable position	File name end flag	1	File name end flag , value is 0x00.

Send package "return tag" should be set to 0x01. Receive and process the return packet.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.3.2 Write file data (CMD = 0x32)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Data length	2	The length of the write data (bytes). Low byte first
0x0002	File data	Variable-length	The data to be written

Note: Each write data don't over 512 bytes, if the file is larger than 512 bytes, according to the order in multiple writing file data.

Send package "return tag" should be set to 0x01. Receive and process the return packet.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.3.3 Close file (CMD = 0x33)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	File checksum	2	The entire file data checksum. Low byte first

Send package "return tag" should be set to 0x01. Receive and process the return packet.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.4 Quick upload file or program

The program files are BIOS file and APP file.

When you update program, you must first confirm that the source of the documents safe, reliable, and formatted correctly, and version is expected

version.

If there is no clear place, do not perform update operation procedures.

1.2.4.1 Open file (CMD = 0x50)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Read/Write mode	1	'r': Read 'w': Write
0x0001	Read/Write type	1	'b': binary 't': Text mode (Now don't support)
0x0002	Hour	1	File time "Hour"
0x0003	Minute	1	File time "Minute"
0x0004	Second	1	File time "Second"
0x0005	Year	1	File time "Year", 2-digit number. For example 0x09 means 2009
0x0006	Month	1	File time "Month"
0x0007	Day	1	File time "Day"
0x0008	Reserved	1	Fill 0
0x0009	File size	4	File size. Low byte first.
0x000d	File name and file extensions	Variable-length	For example: "abc.txt"
Variable position	File name end flag	1	File name end flag , value is 0x00.

Send package "return tag" should be set to 0x01. Receive and process the return packet.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.4.2 Write file data (CMD = 0x51)

Send Packet

Data position	Data Items	Length(byte)	Description
0x0000	Data length	2	The length of the write data (bytes). Low byte first
0x0002	Data block sequence number	2	Numbered starting from 0. Low byte first.
0x0004	Data block length	2	The data block length (in bytes). Low byte first.. Each time a file is updated, according to a fixed length data block, and then the block to send. For all packets, the data block length value should be the same.
0x0006	File data	Variable-length	The data to be written

Note: Each write data don't over 512 bytes, if the file is larger than 512 bytes, according to the order in multiple writing file data.

Send package "return tag" can be set to 0x00 or 0x01. When it be set to 0x00, no return packet. You can continue to send the next packet.

Return Packet (According to the "return tag" to determine)

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x00 Failed ; 0x01 Success

1.2.4.3 Close file (CMD = 0x52)

Send Packet

Data position	Data Items	Length(byte)	Description
0x0000	Total number of data blocks	2	The entire file into a number of data blocks. Low byte first.

Send package "return tag" identification file type, set according to the following table:

- 0x01 : General file.
- 0x02 : Reserved
- 0x03 : BIOS program file
- 0x04 : APP program file

Send package "return tag" should be set to 0x01. Receive and process the return packet.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Number of error block	1	0: No error. >0 and <= 36: The number of error packets. <i>The number of erroneous data returned does not mean that all of the error data segment, You can be re-issued this known error data segment, and then receive the new results, to know there is no error so far.</i>
0x0001	Error block sequence number	Variable-length	Each error packet sequence number 2 bytes, Low byte first. The data length is: the number of error packets * 2

1.2.5 Time query and set (CMD = 0x47)

1.2.5.1 Query time

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x01	>=1	0x01: Query time.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x01	1	0x01: Query time.
0x0001	Time info	≥ 7	Byte 1 : Second Byte 2 : Minute Byte 3 : Hour Byte 4 : Week (0: Sunday, 1: Monday,...) Byte 5 : Day Byte 6 : Month Byte 7 : Year(2-digit number, for example: 0x09 means 2009)

1.2.5.2 Set time

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x00	1	0x00: Set time
0x0001	Time info	≥ 7	Byte 1 : Second Byte 2 : Minute Byte 3 : Hour Byte 4 : Week (0: Sunday, 1: Monday,...) Byte 5 : Day Byte 6 : Month Byte 7 : Year(2-digit number, for example: 0x09 means 2009)

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0 or 1	≥ 1	Byte 1 : 0 Failed; 1 Success. Other: Ignore

1.2.6 Brightness Control (CMD = 0x46)

1.2.6.1 Query Brightness (CMD = 0x46)

Send data

Data position	Data Items	Length(byte)	Description
0x0000	0x01	1	0x01: Query the brightness

Return data

Data position	Data Items	Length(byte)	Description
0x0000	1	1	0x01: Query the brightness
0x0001	Brightness value	24	24 bytes for 24 hours brightness value. invalid value is 0~31, the value large than 31 means auto detect and control by sensor Byte 1: value of 0:00~0:59 Byte 2: value of 1:00~1:59 ... Byte 1: value of 23:00~23:59

1.2.6.2 Set Brightness (CMD = 0x46)

Send data

Data position	Data Items	Length(byte)	Description
0x0000	0	1	0x00 : Set brightness
0x0001	Brightness value	24	24 bytes for 24 hours brightness value. invalid value is 0~31, the value large than 31 means auto detect and control by sensor Byte 1: value of 0:00~0:59 Byte 2: value of 1:00~1:59 ... Byte 1: value of 23:00~23:59

Return data

Data position	Data Items	Length(byte)	Description
0x0000	0 or 1	>=1	Byte 1: 0 failed, 1success Other: Ignore

1.2.7 Query version info (CMD = 0x2e)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x00	1	Reserved

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x00	1	Reserved
0x0001	Card type code	1	Controller type code
0x0002	Logic version	1	Bit0~3: The minor version. Bit4~7: The major version.
0x0003	Bios version	1	Bit0~3: The minor version. Bit4~7: The major version.
0x0004	Reserved	1	
0x0005	Reserved	1	
0x0006	Reserved	1	
0x0007	APP version	1	Bit0~3: The minor version. Bit4~7: The major version.
0x0008	Reserved	Variable-length	

1.2.8 Power ON/OFF info (CMD = 0x45)**1.2.8.1 Query power ON/OFF info (CMD = 0x45)**

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x01	1	0x01: Query power ON/OFF info.

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	1	1	0x01: Query power ON/OFF info.
0x0001	Time value	4	Byte 1~2 : The hour and minute of "ON" time. Byte 3~4 : The hour and minute of "OFF" time. When "On", "off" time is the same that has been "On"; if hour > 23 or minutes > 59 indicates that the time is invalid.
0x0005	Reserved	2	Set to 0

1.2.8.2 Set power ON/OFF info (CMD = 0x45)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0	1	0x00: Set power ON/OFF info.
0x0001	Time value	4	Byte 1~2 : The hour and minute of "ON" time. Byte 3~4 : The hour and minute of "OFF" time. When "On", "off" time is the same that has been "On"; if hour > 23 or minutes > 59 indicates that the time is invalid.
0x0005	Reserved	2	Set to 0

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0 or 1	>=1	Byte 1 : 0 failed, 1 success. Other: Ignore

1.2.9 Power ON/OFF control (CMD = 0x76)

1.2.9.1 Query power ON/OFF info (CMD = 0x76)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0x01	1	0x01: Query software power ON/OFF info.

For example : a5 68 32 01 76 01 01 13 01 ae

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	1	1	0x01: Query software power ON/OFF info.
0x0001	Power status	1	
0x0002	Value	8	Byte 1~2: The hour and minute of "ON" time Byte 3~4: The hour and minute of "OFF" time Byte 5~8: Reserved

1.2.9.2 Set power ON/OFF (CMD = 0x76)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0	1	0x00 : Set software power ON/OFF
0x0001	Switch command	1	0: Immediately turn off the screen. 1: Immediately open the screen.
0x0002	Value	8	Fill 0.

Immediately turn off the screen instance : a5 68 32 01 76 01 00 00 00 00 00 00 00 00 00 12 01 ae

Immediately open the screen instance : a5 68 32 01 76 01 00 01 00 00 00 00 00 00 00 13 01 ae

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	0	1	Set software power ON/OFF
0x0001	Status	>=1	Byte 1 : 0: Immediately turn off the screen. 1: Immediately open the screen. Other: Ignore

1.2.10 Query temperature/humidity (CMD = 0x75)

1.2.10.1 Query temperature, humidity

This protocol requires the support of a new APP version.

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Query flag	>=1	Byte 1 : Bit0: Do you want to query temperature? (0 No,1 Yes) Bit1: Do you want to query humidity? (0 No,1Yes)

			other : Reserved

Return Packet:

There are two returned data format, when the returned data length is 8 bytes, the following meanings:

Data position	Data Items	Length(byte)	Description
0x0000	Query flag	1	Meaning with sent packets
0x0001	Degrees Celsius	2	Temperature value(Degrees Celsius) : Byte 1 : Bit7: numerical symbols. 1 negative, 0 positive. Bit6~0: The high 7 bits in the integer part of the temperature absolute value. Byte 2 : Bit7~4: The low 4 bits in the integer part of the temperature absolute value. Bit3~0: The fractional part. its unit is 1/16 (0.0625).
0x0003	Fahrenheit	2	Temperature value(Fahrenheit) :
0x0009	Temperature adjusted value	1	Bit7: 1 Fahrenheit, 0 Degrees Celsius Bit6: 1 negative, 1 positive Bit5~0: The absolute value of the temperature adjustment.
0x000a	Humidity values	1	Humidity value。 Valid value :0~100
0x000d	Humidity adjustment value	1	Bit7: Reserved Bit6: 1 negative, 1 positive Bit5~0: The absolute value of the humidity adjustment.

When the returned data length is 14 bytes, the following meanings:

Data position	Data Items	Length(byte)	Description
0x0000	Query flag	1	Meaning with sent packets
0x0001	Degrees Celsius	4	Temperature value (Degrees Celsius) string. For example:"23","-3"
0x0005	Fahrenheit	4	Temperature value (Fahrenheit) string. For example:"96"

0x0009	Temperature adjusted value	1	Bit7: 1 Fahrenheit, 0 Degrees Celsius Bit6: 1 negative, 1 positive Bit5~0: The absolute value of the temperature adjustment.
0x000a	Humidity values	3	Humidity value string. For example: "80"
0x000d	Humidity adjustment value	1	Bit7: Reserved Bit6: 1 negative, 1 positive Bit5~0: The absolute value of the humidity adjustment.

1.2.11 Remove file (CMD = 0x2c)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	File name and file extensions	Variable-length	For example: "abc.txt"
Variable position	File name end flag	1	Fill name end flag. Value is 0x00

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Result	1	0x01 Success; Others Failed

1.2.12 Query free disk space (CMD = 0x29)

Send Packet:

Data position	Data Items	Length(byte)	Description
0x0000	1 or 0	1	1 : Disk "User" 0 : Disk "System"

Return Packet:

Data position	Data Items	Length(byte)	Description
0x0000	Return value	1	>0 Success; Others failed.
0x0001	The size of free disk space	4	Low byte first.